

PS-AFTER-MID — Weeks 9–15 · 12 Oct – 29 Nov 2026 · FINAL (W15)

Scope: Weeks 9–15 (12 Oct – 29 Nov 2026) — sampling distributions & CLT, hypothesis testing I (z & t), hypothesis testing II (chi-square & Type I/II), confidence intervals, regression & correlation, ending at the Week 15 final. **Siblings:** PS-NAV.md (master map) · PS-BEFORE-MID.md (Weeks 1–8). Week-colored headers below match the NAV schedule. **Exam split:** 60% numerical + 40% theory. Hypothesis testing is the single most important topic — one 15-mark question guaranteed. **Exam order:** numerals → theory (final). For hypothesis testing write the 4 steps explicitly — correct structure earns 60% of marks.

At a glance (W9–W15)

W	Dates	Variant	File	Topics	Chapter
W9	12–18 Oct	P1	AFTER	Sampling distributions & CLT (recovery)	06
W10	19–25 Oct	P0	AFTER	Hypothesis testing I: z-test & t-test	07
W11	26 Oct–01 Nov	P0	AFTER	Hypothesis testing II: chi-square & Type I/II	07
W12	02–08 Nov	P0	AFTER	Confidence intervals	07
W13	09–15 Nov	P1	AFTER	Regression & correlation	08
W14	16–22 Nov	P0	AFTER	Final taper — past paper FIRST	01–08
W15	23–29 Nov	FINAL	AFTER	Exam — execute	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week.

W9 — Sampling Distributions & CLT · 12–18 Oct · P1 · 5 hrs

Banner: Rotation PS (Mon+Tue) · Tier P1 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
Recovery week — ledger MUST clear to 0.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 9	lines 158–179
Fear-Killer Pack	Fear-Killer-Packs.md → Week 5	lines 44–49
Week manifest	weeks/PS-W9.md	full file
Chapter breakdown	03-Chapter-Breakdowns/06-Sampling-and-Sampling-Distributions.md	embedded below

Definitions (verbatim)

Term	Definition
Population	The complete set of all elements of interest.
Sample	A subset of the population selected for study.
Parameter	A numerical characteristic of a population (μ , σ^2).
Statistic	A numerical characteristic of a sample ($\bar{x}$, s^2).
Standard error	The standard deviation of a sampling distribution, $\sigma/\sqrt{n}$.
Central Limit Theorem	For a random sample of size n from any population with mean μ and finite variance σ^2 , the sampling distribution of $\bar{x}$ approaches $N(\mu, \sigma/\sqrt{n})$ as $n \rightarrow \infty$.
Sampling error	The discrepancy between a statistic and the corresponding parameter due to sampling.

Formulas (verbatim)

Formula	Statement
Standard error of mean	$SE = \sigma / \sqrt{n}$
With FPC (w/o replacement)	$SE = (\sigma/\sqrt{n}) \times \sqrt{((N-n)/(N-1))}$
Sampling distribution	$X \sim N(\mu, \sigma^2/n)$ (n large, CLT)
Z for sample mean	$Z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #10 Central Limit Theorem illustration
Numericals	Numerical-Book.md #57 standard error = $\sigma/\sqrt{n}$ · #58 CLT probability of sample mean in range

Books · Chapters · Media

Resource	Where
Devore	Ch 5 (sampling, CLT)
StatQuest	CLT

Fear-Killer Pack Week 5 (verbatim)

Resources: StatQuest (CLT) • Brandon Foltz (Sampling) • Devore Ch.6-7

1. A population has mean $\mu = 100$ and standard deviation $\sigma = 20$. Draw samples of size $n = 36$. State the sampling distribution of the sample mean (shape, mean, standard error). Calculate $P(95 < X < 105)$. Now for $n = 100$, recalculate the standard error and probability. How does increasing n affect the sampling distribution?
2. Compare stratified sampling, cluster sampling, and systematic sampling. A university has 10,000 students across 5 faculties: Engineering (3000), Science (2500), Arts (2000), Business (1500), Law (1000). You need a sample of 500 students. Show your sampling plan for each method. Which method gives the most representative sample and why?
3. State the Central Limit Theorem mathematically. A factory produces bolts with mean diameter 10mm and variance 4mm². For a sample of 40 bolts, what is $P(\text{sample mean} > 10.5\text{mm})$? What if the population is heavily skewed — does CLT still apply and at what sample size? Show the derivation and the Z-score.

Narrative — Week 9 (verbatim)

Topics: Population vs sample; Parameter vs statistic; Sampling distribution of the mean; Standard error; Central Limit Theorem; Sampling methods (simple random, stratified, cluster, systematic)

Recovery note: If midterm scored below 85%, take 2 days at reduced load (20 hrs/week total). Recalibrate — a 78% midterm with 95%+ finals still yields B+/A-.

Resources: Devore Ch.5; StatQuest — CLT

Practice: - CLT: for any population with mean μ and variance σ^2 , the sampling distribution of $\bar{x}$ approaches $N(\mu, \sigma/\sqrt{n})$ as $n \rightarrow \infty$ — retrieve from memory - **Trap:** CLT applies to the sampling distribution of the mean, NOT to the population distribution - Standard error = $\sigma/\sqrt{n}$ — derive from first principles - **Interleaved retrieval (20 min):** Bayes theorem — 3 problems from blank page (medical testing, spam detection). Then compute mean and variance from raw data. All closed book.

Retrieval protocol: Every practice item starts from a blank page. No peeking at notes.

Deliverable: 5 CLT problems (probability of sample mean in a range)

Time budget: 5 hrs

Chapter 6 — Sampling & Sampling Distributions (verbatim)

Weight: ★★☆☆☆

Difficulty: ★★☆☆☆

Importance: ★★★★★

Learning Objectives

Explain

- Population, parameter & statistic
- Objects of sampling
- Sampling & non-sampling errors
- Central Limit Theorem

Calculate

- Sampling distribution of the mean
- Standard errors

Apply

- Random sampling with & without replacement
- Sequential sampling

Differentiate

- Simple random, stratified, cluster, systematic sampling

Must Memorize

Population: The complete set of all elements of interest.

Sample: A subset of the population selected for study.

Parameter: A numerical characteristic of a population (μ , σ^2).

Statistic: A numerical characteristic of a sample ($\bar{x}$, s^2).

Standard error: The standard deviation of a sampling distribution, $\sigma/\sqrt{n}$.

Central Limit Theorem: For a random sample of size n from any population with mean μ and finite variance σ^2 , the sampling distribution of $\bar{x}$ approaches $N(\mu, \sigma/\sqrt{n})$ as $n \rightarrow \infty$.

Sampling error: The discrepancy between a statistic and the corresponding parameter due to sampling.

Must Understand

- CLT applies to the sampling distribution of the mean, NOT the population
- Standard error shrinks as n grows: $SE = \sigma/\sqrt{n}$
- Stratified sampling divides the population into homogeneous strata for representativeness
- Sampling error (chance) vs non-sampling errors (bias, measurement, non-response)
- With vs without replacement: without replacement uses the finite population correction

Must Practice

Population $\mu = 100$, $\sigma = 20$

$n = 36$: $\bar{X} \sim N(100, 20/\sqrt{36} = 3.33)$

$P(95 < \bar{X} < 105) \rightarrow Z = \pm 1.5 \rightarrow \approx 0.8664$

$n = 100$: $SE = 2$, $P(95 < \bar{X} < 105) \rightarrow Z = \pm 2.5 \rightarrow \approx 0.9876$

Common Mistakes

1. **Applying CLT to the population distribution** — it describes $\bar{x}$, not X .
2. **SE vs σ** — standard error is $\sigma/\sqrt{n}$, always smaller for $n > 1$.
3. **Ignoring finite population correction** for sampling without replacement.
4. **Confusing parameter (population) with statistic (sample)** — μ vs $\bar{x}$.
5. **Conflating sampling and non-sampling errors.**

Typical Questions

- 1. State CLT mathematically and apply to a bolt-diameter example.
- 2. Compute $P(95 < X < 105)$ for $n = 36$ and $n = 100$; comment on n .
- 3. Compare stratified, cluster, systematic sampling on a university dataset.
- 4. Define standard error; how does it change with n ?
- 5. Distinguish sampling vs non-sampling errors with examples.

Examiner Expectations

Level	Performance
Pass	Defines population/parameter/statistic, states CLT
Good	Computes standard error and probabilities for $\bar{x}$
Excellent	Compares sampling methods, applies finite population correction
Full marks	CLT problems and sampling plans solved correctly

Formula Sheet

Standard error of mean: $SE = \sigma / \sqrt{n}$
With FPC (w/o replacement): $SE = (\sigma/\sqrt{n}) \times \sqrt{((N-n)/(N-1))}$
Sampling distribution: $\bar{X} \sim N(\mu, \sigma^2/n)$ (n large, CLT)
Z for sample mean: $Z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$

GPA Priority: ■ Must Win (CLT guaranteed theory + numerical)

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 SE + probability + Q3 CLT) · def/formula skim (SE + CLT) · same-problem drill #57, #58
- **Same-problem drill target:** SE < 30 s; P(sample mean in range) < 2 min
- **Deliverable:** 5 CLT problems (probability of sample mean in a range)
- **Trap:** CLT applies to the sampling distribution of the mean, NOT the population. Recovery mandate: ledger MUST clear to 0.

W10 — Hypothesis Testing I: z-test & t-test · 19–25 Oct · P0 · 6 hrs

Banner: Rotation PS (Mon+Tue) · Tier P0 · Time budget 6 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 10	lines 180–205
Fear-Killer Pack	Fear-Killer-Packs.md → Week 6	lines 51–57 (Q1–Q3)
Week manifest	weeks/PS-W10.md	full file
Chapter breakdown	03-Chapter-Breakdowns/07-Statistical-Inference-and-Testing-of-Hypothesis.md	embedded below

Definitions (verbatim)

Term	Definition
Point estimate	A single value used to estimate a population parameter.
Confidence interval	An interval that contains the parameter with a stated confidence level, $(1-\alpha)100\%$.
Null hypothesis H_0	The statement being tested, usually of "no effect" or "no difference".
Alternative hypothesis H_1	The statement we accept if H_0 is rejected.
Type I error (α)	Rejecting a true H_0 .

Type II error (β)	Failing to reject a false H_0 .
p-value	The probability of obtaining a test statistic as extreme as observed, assuming H_0 is true.

Formulas (verbatim)

Formula	Statement
z-test (σ known)	$z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$
t-test (σ unknown)	$t = (\bar{x} - \mu) / (s/\sqrt{n}), df = n-1$
Two-sample t	$t = (\bar{x}_1-\bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
One-proportion z	$z = (\hat{p} - p_0) / \sqrt{(p_0(1-p_0)/n)}$
Decision	reject H_0 if $ \text{test statistic} > \text{critical value}$, or if $p\text{-value} < \alpha$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #4 one-tailed vs two-tailed rejection regions · #5 Type I and Type II error diagram
Numericals	Numerical-Book.md #59 z-test · #60 t-test · #61 critical value from table · #62 p-value interpretation

Books · Chapters · Media

Resource	Where
Devore	Ch 8 (hypothesis testing)
StatQuest	Hypothesis Testing

Fear-Killer Pack Week 6 (verbatim — Q1–Q3)

Resources: StatQuest (Hypothesis Testing) • Devore Ch.8-9

- A coffee machine dispenses 200ml per cup on average. A sample of 25 cups gives mean = 195ml, s = 10ml. Test at $\alpha = 0.05$ whether the machine is underfilling. State H_0 and H_1 , choose one-tailed or two-tailed, calculate the t-statistic, find the critical value, and state your conclusion. Then calculate the p-value. Would the conclusion change at $\alpha = 0.01$?
- 100 light bulbs from Brand A have mean life 1200 hours, s = 80 hours. 100 bulbs from Brand B have mean life 1150 hours, s = 90 hours. Test at $\alpha = 0.05$ whether Brand A lasts longer than Brand B. Use a two-sample t-test. State H_0 , H_1 , calculate the test statistic, degrees of freedom, and conclude.
- A survey of 200 people shows 120 prefer Product A over Product B. Test at $\alpha = 0.05$ whether more than 50% prefer Product A. Use a one-proportion z-test. Calculate the test statistic and p-value. Also construct a 95% confidence interval for the true proportion. Does the CI support the test conclusion?

Narrative — Week 10 (verbatim)

Topics: Null hypothesis (H_0); Alternative hypothesis (H_1); Test statistic; Critical region; Type I error (α); Type II error (β); p-value; One-tailed vs two-tailed tests

Resources: Devore Ch.8; StatQuest — Hypothesis Testing

Practice: - **The 4-step procedure (retrieve from memory before looking at notes):** 1. State H_0 and H_1 2. Choose test statistic ($z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$ for known σ ; $t = \text{same formula but with } s \text{ for unknown } \sigma$) 3. Find critical value (z_{α} or t_{α} from table) 4. Decision: reject H_0 if $|\text{test statistic}| > \text{critical value}$, or if $p\text{-value} < \alpha$ - **Decision flowchart (retrieve from blank page):** - σ known $\rightarrow$ z-test - σ unknown $\rightarrow$ t-test - One sample vs two sample $\rightarrow$ different formulas - **Interleaved retrieval (20 min):** Distribution identification — given 5 scenarios, name the distribution and parameters. Then 3 Bayes problems. All closed book.

Killer trap: One-tailed vs two-tailed. If H_1 says "greater than" it is one-tailed (α in one tail). If H_1 says "different from" it is two-tailed ($\alpha/2$ in each tail). This is the #1 exam mistake.

Deliverable: 10 hypothesis testing problems (z-test and t-test, one-tailed and two-tailed) — all from blank page

Time budget: 6 hrs

Chapter 7 — Statistical Inference & Testing of Hypothesis (verbatim)

Difficulty: ★★☆☆☆

Importance: ★★★★★

Learning Objectives

Explain

- Estimation (point & interval)
- Tests of hypothesis
- Type I / Type II errors, power
- One-tail vs two-tail tests

Calculate

- Confidence intervals (mean z & t, proportion)
- Sample size for desired margin of error
- Test statistics (z, t, χ^2)
- Chi-square test for independence / goodness of fit

Apply

- The 4-step hypothesis-testing procedure

Must Memorize

Point estimate: A single value used to estimate a population parameter.

Confidence interval: An interval that contains the parameter with a stated confidence level, $(1-\alpha)100\%$.

Null hypothesis H_0 : The statement being tested, usually of "no effect" or "no difference".

Alternative hypothesis H_1 : The statement we accept if H_0 is rejected.

Type I error (α): Rejecting a true H_0 .

Type II error (β): Failing to reject a false H_0 .

p-value: The probability of obtaining a test statistic as extreme as observed, assuming H_0 is true.

Chi-square test: $\Sigma(O-E)^2/E$ comparing observed with expected frequencies.

Must Understand

- "95% confidence" means 95 of 100 such intervals contain the parameter — NOT a 95% probability the parameter lies in this interval
- One-tailed vs two-tailed: H_1 "greater than" $\rightarrow \alpha$ in one tail; "different from" $\rightarrow \alpha/2$ in each tail
- σ known $\rightarrow z$; σ unknown $\rightarrow t$
- $p < \alpha \rightarrow$ reject H_0 (assuming α stated)
- χ^2 expected frequency = (row total $\times$ column total)/grand total
- Type II / power are rarely computed in class but asked as concepts

Must Practice

$H_0: \mu = 200, H_1: \mu < 200$ (one-tailed)

$\bar{x} = 195, s = 10, n = 25$

$t = (195-200) / (10/\sqrt{25}) = -2.5$

$t_{\text{crit}}(0.05, df=24) \approx -1.711 \rightarrow$ reject H_0 ; $p < 0.05$

Brand A: $\bar{x}_1=1200, s_1=80, n_1=100$; Brand B: $\bar{x}_2=1150, s_2=90, n_2=100$

$H_0: \mu_1 = \mu_2, H_1: \mu_1 > \mu_2 \rightarrow$ compute t , compare with t_{crit}

Common Mistakes

1. **One vs two-tailed critical value** — halve α for two-tailed.
2. **σ known/unknown mix-up** — z-test requires known σ .
3. **Misreading p-value** — always conditional on H_0 being true.
4. **CI interpretation** — the parameter is fixed; the interval varies.
5. **χ^2 expected counts** — must use row $\times$ col/grand total, not averages.
6. **t vs z at small n** — use t, $df = n-1$.

Typical Questions

1. Coffee-machine underfill test at $\alpha = 0.05$ and $\alpha = 0.01$ (one-tailed t).
2. Two-sample t: which brand lasts longer?
3. One-proportion z-test + 95% CI; does the CI agree with the test?
4. Chi-square test for independence on a 2 $\times$ 3 table.
5. Explain what 95% confidence actually means (100 intervals example).

Examiner Expectations

Level	Performance
Pass	States H_0/H_1 , computes test statistic
Good	Chooses correct test, finds critical value, concludes
Excellent	p-values, CI agreement, chi-square, Type I/II concepts
Full marks	Complete 4-step solutions under time pressure, 15-mark answer

Formula Sheet

z-test (σ known): $z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$
t-test (σ unknown): $t = (\bar{x} - \mu) / (s/\sqrt{n})$, $df = n-1$
Two-sample t: $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
One-proportion z: $z = (\hat{p} - p_0) / \sqrt{(p_0(1-p_0)/n)}$
CI for μ (σ known): $\bar{x} \pm z_{\alpha/2} \sigma/\sqrt{n}$
CI for μ (unknown): $\bar{x} \pm t_{\alpha/2} s/\sqrt{n}$
CI for proportion: $\hat{p} \pm z_{\alpha/2} \sqrt{(\hat{p}(1-\hat{p})/n)}$
Sample size: $n = (z_{\alpha/2} \sigma / E)^2$
Chi-square: $\chi^2 = \sum (O-E)^2 / E$, $df = (r-1)(c-1)$

GPA Priority: ■ Must Win (15-mark question guaranteed)

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 coffee-machine t-test + Q3 one-proportion) · def/formula skim (H_0/H_1 /Type I/II + z/t formulas) · same-problem drill #59, #60
- **Same-problem drill target:** 4-step procedure < 20 min/problem; choose z vs t instantly
- **Deliverable:** 10 hypothesis testing problems (z-test and t-test, one-tailed and two-tailed) — all from blank page
- **Trap:** one-tailed vs two-tailed — α in one tail vs $\alpha/2$ in each tail (#1 exam mistake)

W11 — Hypothesis Testing II: Chi-Square & Type I/II · 26 Oct–01 Nov · P0 · 5 hrs

Banner: Rotation PS (Mon+Tue) · Tier P0 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 11	lines 206–225
Fear-Killer Pack	Fear-Killer-Packs.md → Week 6	lines 51–57 (Q4)
Week manifest	weeks/PS-W11.md	full file
Chapter breakdown	03-Chapter-Breakdowns/07-... 7 / 16	reference (embedded W10)

Definitions (verbatim)

Term	Definition
Type I error (α)	Rejecting a true H_0 .
Type II error (β)	Failing to reject a false H_0 .
Power of a test	Power = $1 - \beta$.
p-value	The probability of obtaining a test statistic as extreme as observed, assuming H_0 is true.
Chi-square test	$\Sigma(O-E)^2/E$ comparing observed with expected frequencies.

Formulas (verbatim)

Formula	Statement
Chi-square	$\chi^2 = \Sigma (O - E)^2 / E$
Expected frequency	(row total $\times$ column total) / grand total
Degrees of freedom	$df = (r - 1)(c - 1)$
Power	Power = $1 - \beta$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #5 Type I and Type II error diagram
Numericals	Numerical-Book.md #63 chi-square O–E computation

Books · Chapters · Media

Resource	Where
Devore	Ch 7–8 (chi-square, Type I/II)
StatQuest	Chi-square test

Fear-Killer Pack Week 6 (verbatim — Q4)

Resources: StatQuest (Hypothesis Testing) · Devore Ch.8-9

4. A chi-square test for independence: 300 people classified by gender (M/F) and product preference (A/B/C). Observed counts: M/A=40, M/B=35, M/C=25, F/A=60, F/B=65, F/C=75. Calculate expected frequencies assuming independence, compute χ^2 , find the critical value at $\alpha = 0.05$ with appropriate degrees of freedom. Are gender and preference independent?

Narrative — Week 11 (verbatim)

Topics: Chi-square test for goodness of fit; Chi-square test for independence; Type I error (rejecting true H_0); Type II error (failing to reject false H_0); Power of a test

Resources: Devore Ch.7-8; StatQuest — Chi-square test

Practice: - $\chi^2 = \Sigma(O - E)^2 / E$ — retrieve formula from memory - Goodness of fit: compare observed frequencies to expected frequencies — closed book - Test for independence: contingency table, expected frequency = (row total $\times$ column total) / grand total — from blank page - Type I vs Type II: $\alpha = P(\text{Type I error})$, $\beta = P(\text{Type II error})$. Power = $1 - \beta$ — retrieve from memory -

Trap: p-value interpretation — "p < 0.05 means reject H_0 " must include "assuming $\alpha = 0.05$ " - **Interleaved retrieval (20 min):** Hypothesis testing 4-step procedure — 2 complete problems (z-test and t-test) from blank page. All closed book.

Deliverable: 5 chi-square problems + 5 Type I/II error identification problems

Time budget: 5 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q4 chi-square independence) · def/formula skim (Type I/II, χ^2) · same-problem drill #63
- **Same-problem drill target:** χ^2 expected counts < 1 min; $df = (r-1)(c-1)$ instant
- **Deliverable:** 5 chi-square problems + 5 Type I/II error identification problems
- **Trap:** χ^2 expected = (row $\times$ col)/grand total, not averages; $p \leq 0.05$ means reject H_0 **assuming $\alpha = 0.05$**

Banner: Rotation PS (Mon+Tue) · Tier P0 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
Start sleep banking: 9 hrs (bedtime 22:00) for 7 nights.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 12	lines 226–246
Fear-Killer Pack	Fear-Killer-Packs.md → Week 7	lines 59–64
Week manifest	weeks/PS-W12.md	full file
Chapter breakdown	03-Chapter-Breakdowns/07-... (CI part)	reference (embedded W10)

Definitions (verbatim)

Term	Definition
Point estimate	A single value used to estimate a population parameter.
Confidence interval	An interval that contains the parameter with a stated confidence level, $(1-\alpha)100\%$.
Margin of error	Half the width of the confidence interval, $z_{\alpha/2} \times \sigma/\sqrt{n}$ (mean) or $z_{\alpha/2} \times \sqrt{\hat{p}(1-\hat{p})/n}$ (proportion).

Formulas (verbatim)

Formula	Statement
CI for μ (σ known)	$\bar{x} \pm z_{\alpha/2} \times \sigma/\sqrt{n}$
CI for μ (σ unknown)	$\bar{x} \pm t_{\alpha/2} \times s/\sqrt{n}$
CI for proportion	$\hat{p} \pm z_{\alpha/2} \times \sqrt{(\hat{p}(1-\hat{p})/n)}$
Sample Size	$n = (z_{\alpha/2} \times \sigma / E)^2$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #7 confidence interval visualization
Numericals	Numerical-Book.md #64 CI for mean (σ known, σ unknown) · #65 CI for proportion · #66 sample size for desired margin of error

Books · Chapters · Media

Resource	Where
Devore	Ch 7 (confidence intervals)
StatQuest	Confidence Intervals

Fear-Killer Pack Week 7 (verbatim)

Resources: StatQuest (Confidence Intervals) • Devore Ch.7-8

- A sample of 16 batteries has mean life 50 hours, $s = 4$ hours. Construct 95% CI for the population mean (σ unknown, use t -distribution). Now σ is known to be 4, recalculate using z . Compare the widths of the two intervals. What sample size is needed to reduce the CI width to ± 1 hour? Calculate for both known and unknown σ .
- A survey of 500 voters finds 280 support Candidate A. Construct 95% CI for the true proportion. Calculate the margin of error. How many voters need to be surveyed to reduce the margin of error to $\pm 2\%$? Use $\hat{p} = 0.5$ (conservative) and $\hat{p} = 0.56$ (from pilot) — what sample sizes result?
- Explain what "95% confidence" actually means. Given 100 different 95% CIs each from independent samples, how many would you expect to contain the true population mean? Show a concrete simulation example. Differentiate this from saying "there's a 95% probability the parameter lies in this interval" — why is that statement incorrect?

Narrative — Week 12 (verbatim)

Topics: Point estimation vs interval estimation; Confidence interval for mean (z and t); Confidence interval for proportion; Sample size determination; Margin of error

Resources: Devore Ch.7; StatQuest — Confidence Intervals

Sleep banking: Begin sleeping 9 hrs (bedtime 22:00) for 7 nights leading to finals. Formula recall degrades 15-20% per lost sleep hour — protect your hypothesis testing and CI formulas.

Practice: - CI for μ (σ known): $\bar{x} \pm z_{\alpha/2} \times \sigma/\sqrt{n}$ — retrieve formula from memory - CI for μ (σ unknown): $\bar{x} \pm t_{\alpha/2} \times s/\sqrt{n}$ — retrieve formula from memory - CI for proportion: $\hat{p} \pm z_{\alpha/2} \times \sqrt{(\hat{p}(1-\hat{p})/n)}$ — retrieve from memory - Sample size for desired margin of error: $n = (z_{\alpha/2} \times \sigma / E)^2$ — retrieve from memory - **Interleaved retrieval (20 min):** 4-step hypothesis testing — 2 complete problems (chi-square test for independence, z-test). Then 3 Bayes problems. All closed book, from blank page.

Deliverable: 10 confidence interval problems (mean, proportion, sample size)

Time budget: 5 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 battery CI + Q2 voter CI) · def/formula skim (3 CI formulas + sample size) · same-problem drill #64, #65, #66
- **Same-problem drill target:** CI < 2 min each; sample size $n = (z \cdot \sigma / E)^2$ instant
- **Deliverable:** 10 confidence interval problems (mean, proportion, sample size)
- **Trap:** "95% confidence" = 95 of 100 intervals contain the parameter — NOT a 95% probability the parameter lies in this interval; z vs t by known/unknown σ

W13 — Regression & Correlation · 09–15 Nov · P1 · 5 hrs

Banner: Rotation PS (Mon+Tue) · Tier P1 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 13	lines 247–267
Fear-Killer Pack	Fear-Killer-Packs.md → Week 8	lines 66–73
Week manifest	weeks/PS-W13.md	full file
Chapter breakdown	03-Chapter-Breakdowns/08-Simple-Regression-and-Correlation.md	embedded below

Definitions (verbatim)

Term	Definition
Correlation coefficient (r)	A measure of the strength and direction of the linear relationship between two variables, $r \in [-1, 1]$.
Regression line	The line of best fit minimizing the sum of squared vertical deviations: $Y = a + bX$.
Least squares	Choosing a and b to minimize $\Sigma(y - a - bx)^2$.
Coefficient of determination (R^2)	The proportion of variance in Y explained by X; $R^2 = r^2$.
Spearman's rank correlation (ρ)	Correlation computed on ranks, $\rho = 1 - 6\Sigma d^2/(n(n^2-1))$.
Probable Error (P.E)	$P.E = 0.6745 \times (1 - r^2)^{1/2}/n$; the interval $r \pm P.E$ assesses significance of correlation.

Formulas (verbatim)

Formula	Statement
Correlation	$r = \Sigma(x-\bar{x})(y-\bar{y}) / \sqrt{(\Sigma(x-\bar{x})^2 \Sigma(y-\bar{y})^2)}$
Slope	$b = \Sigma(x-\bar{x})(y-\bar{y}) / \Sigma(x-\bar{x})^2$
Intercept	$a = \bar{y} - b \bar{x}$

R^2	r^2
Spearman	$\rho = 1 - 6\sum d^2 / (n(n^2-1))$
Probable Error	$P.E = 0.6745 (1 - r^2) / \sqrt{n}$
Significance	correlation significant if $r \pm P.E$ excludes 0

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #6 scatter plot with regression line
Numericals	Numerical-Book.md #67 b (slope) · #68 a (intercept) · #69 r (correlation) · #70 R^2 interpretation

Books · Chapters · Media

Resource	Where
Devore	Ch 12 (regression)
StatQuest	Linear Regression
Brandon Foltz	Statistics 101 — Regression playlist

Fear-Killer Pack Week 8 (verbatim)

Resources: StatQuest (Regression) • Brandon Foltz (Regression) • Devore Ch.10-11

- Given data: X (hours studied) = [2,4,6,8,10,12], Y (exam score) = [55,60,70,75,85,95]. Calculate the correlation coefficient r , the regression line $Y = a + bX$, and interpret b and a . Predict Y when $X = 7$ and $X = 20$. Which prediction is more reliable and why?
- For the same data, calculate R^2 and interpret it. Then compute the residual for each point. Draw the scatter plot with the regression line and show the residuals. Explain the least squares principle — why minimize vertical distances and not perpendicular distances? Show the derivation of $b = \Sigma((x-\bar{x})(y-\bar{y})) / \Sigma((x-\bar{x})^2)$.
- Rank correlation: Two judges rank 8 contestants as follows: Judge A: [3,1,2,5,4,7,6,8], Judge B: [2,3,1,4,6,5,8,7]. Calculate Spearman's rank correlation coefficient. Interpret the value. What does $\rho = +1, -1$, and 0 mean in terms of agreement between judges?
- Differentiate Y on X vs X on Y regression. Given the same data, compute both regression lines and show they are not inverses of each other. Which line should you use when you want to predict Y from X ? When would you ever need the X on Y line?
- Using $r \approx 0.97$ from Q1 ($n = 6$) and the Spearman ρ from Q3, compute the Probable Error $P.E = 0.6745 \times (1 - r^2)/\sqrt{n}$ for each. State the significance limits $r \pm P.E$ and decide whether each correlation is significant (interval free of zero) or not. When n is small, why does the $P.E$ interval widen, and what does that mean for drawing conclusions about correlation?

Narrative — Week 13 (verbatim)

Topics: Scatter plot; Correlation coefficient (r); Coefficient of determination (R^2); Regression line (least squares); Regression lines: Y on X vs X on Y (they are different)

Resources: Devore Ch.12; StatQuest — Linear Regression

Practice: - Correlation: r ranges from -1 to $+1$. $r = 0$ = no linear correlation — retrieve from memory - R^2 = proportion of variance explained by the model — retrieve from memory - Regression line: $Y = a + bX$ — derive formulas from blank page - $b = \Sigma((x-\bar{x})(y-\bar{y})) / \Sigma((x-\bar{x})^2)$ - $a = \bar{y} - b\bar{x}$ - **Trap:** Regression of Y on X minimizes vertical residuals. Regression of X on Y minimizes horizontal residuals. They give different lines. This is a common exam surprise. - **Interleaved retrieval (20 min):** Timed formula speed drill — write all 5 hypothesis testing formulas, all 3 CI formulas, Bayes theorem, regression formulas from memory in 10 min. Then solve 1 complete hypothesis testing problem. All closed book.

Deliverable: 5 regression problems (compute line, interpret coefficients, predict)

Time budget: 5 hrs

Chapter 8 — Simple Regression & Correlation (verbatim)

Weight: ★★★★★

Difficulty: ★★★★★

Learning Objectives

Explain

- Scatter diagrams
- Least-squares principle

Calculate

- Correlation coefficient (Pearson r)
- Regression lines (Y on X, X on Y)
- Rank correlation (Spearman ρ)
- Probable Error (P.E)
- Coefficient of determination R^2

Interpret

- Slope, intercept, residuals, R^2
- Significance of correlation via P.E

Differentiate

- Y on X vs X on Y regression lines
- Correlation vs causation

Must Memorize

Correlation coefficient (r): A measure of the strength and direction of the linear relationship between two variables, $r \in [-1, 1]$.

Regression line: The line of best fit minimizing the sum of squared vertical deviations: $Y = a + bX$.

Least squares: Choosing a and b to minimize $\sum(y - a - bx)^2$.

Coefficient of determination (R^2): The proportion of variance in Y explained by X; $R^2 = r^2$.

Spearman's rank correlation (ρ): Correlation computed on ranks, $\rho = 1 - 6\sum d^2 / (n(n^2 - 1))$.

Probable Error (P.E): $P.E = 0.6745 \times (1 - r^2) / \sqrt{n}$; the interval $r \pm P.E$ assesses significance of correlation.

Must Understand

- $b = \sum((x - \bar{x})(y - \bar{y})) / \sum((x - \bar{x})^2)$; $a = \bar{y} - b\bar{x}$
- Regression of Y on X minimizes vertical residuals; X on Y minimizes horizontal — they are different lines
- r near ± 1 = strong linear relation; $r = 0$ = none; r says nothing about non-linear or causation
- R^2 = proportion of variance explained
- Spearman uses ranks → robust to outliers, handles ordinal data
- Correlation is significant if the interval $r \pm P.E$ does not contain zero

Must Practice

X (hours): 2, 4, 6, 8, 10, 12 ; Y (score): 55, 60, 70, 75, 85, 95

$b = \sum(x - \bar{x})(y - \bar{y}) / \sum(x - \bar{x})^2 = 4.21$, $a = \bar{y} - b\bar{x} = 42.5$

$Y = 42.5 + 4.21X$; predict $X=7 \rightarrow 71.97$, $X=20 \rightarrow 126.7$ (extrapolation unreliable)

Common Mistakes

1. **Y on X vs X on Y** — they are not inverses; use Y on X to predict Y from X.
2. **Extrapolation** — predictions far outside data range are unreliable.
3. **$r \neq$ causation** — correlation does not imply cause.
4. **R^2 vs r** — $R^2 = r^2$ is proportion of variance explained.

5. **P.E formula** — $0.6745 \times (1-r^2)^{1/4}/n$, not $(1-r)/\sqrt{n}$.
6. **Spearman with tied ranks** — use average ranks.

Typical Questions

1. Compute r , regression line $Y = a + bX$, interpret a and b , predict Y at $X = 7$ and $X = 20$.
2. Calculate R^2 and residuals; explain least squares (why vertical?).
3. Spearman's rank correlation for two judges' rankings.
4. Compute P.E; assess significance of r . Why does the interval widen at small n ?
5. Differentiate Y on X vs X on Y regression with the same data.

Examiner Expectations

Level	Performance
Pass	Computes r and regression coefficients
Good	Predicts, interprets slope and R^2
Excellent	Rank correlation, P.E significance, line-direction nuance
Full marks	Full regression + correlation solutions fast, error-free

Formula Sheet

Correlation:	$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{(\sum (x-\bar{x})^2 \sum (y-\bar{y})^2)}}$
Slope:	$b = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$
Intercept:	$a = \bar{y} - b \bar{x}$
R^2 :	r^2
Spearman:	$\rho = 1 - \frac{6\sum d^2}{(n(n^2-1))}$
Probable Error:	$P.E = 0.6745 (1 - r^2) / \sqrt{n}$
Significance:	correlation significant if $r \pm P.E$ excludes 0

GPA Priority: ☒ **Must Win (numerical, finals regular)**

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 r + line + prediction + Q2 R^2 residuals) · def/formula skim (r , R^2 , a/b) · same-problem drill #67, #68, #69, #70
- **Same-problem drill target:** regression line a , b from blank page < 10 min; r computed and interpreted
- **Deliverable:** 5 regression problems (compute line, interpret coefficients, predict)
- **Trap:** Y on X vs X on Y are different lines — Y on X minimizes vertical residuals; extrapolation beyond the data range is unreliable

W14 — Final Exam Preparation (Taper) · 16–22 Nov · P0 · 8 hrs

Banner: Rotation PS (Mon+Tue) · Tier P0 · Time budget 8 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
START with the past paper, NOT with review.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 14	lines 268–295
Fear-Killer Pack	Fear-Killer-Packs.md → all packs review	lines 11–73
Week manifest	weeks/PS-W14.md	full file
Chapter breakdowns	all 8 files	reference (embedded W1, W2, W3, W5, W6, W9, W10, W13)

Definitions (verbatim) — full W1–W13 sweep

Term	Definition
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Statistics	The science of collecting, organizing, presenting, analyzing and interpreting data to make decisions.
Mean	The sum of all observations divided by their number.
Median	The middle value when data is arranged in order; Q2.
Standard deviation	The positive square root of the variance — the average deviation of observations from the mean.
Conditional probability	$P(A B) = P(A \cap B) / P(B)$, the probability of A given that B has occurred.
Bayes' rule	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$.
Expected value	$E[X] = \sum x_i P(x_i)$ (discrete) or $\int x f(x)dx$ (continuous).
Binomial	$X \sim B(n, p)$ — number of successes in n independent trials, fixed p. $P(X=x) = C(n,x)p^x(1-p)^{n-x}$.
Normal	$X \sim N(\mu, \sigma^2)$; $Z = (X-\mu)/\sigma \sim N(0,1)$; symmetric bell curve.
Standard error	The standard deviation of a sampling distribution, $\sigma/\sqrt{n}$.
Central Limit Theorem	For a random sample of size n from any population with mean μ and finite variance σ^2 , the sampling distribution of $\bar{x}$ approaches $N(\mu, \sigma/\sqrt{n})$ as $n \rightarrow \infty$.
Null hypothesis H_0	The statement being tested, usually of "no effect" or "no difference".
Type I error (α)	Rejecting a true H_0 .
Type II error (β)	Failing to reject a false H_0 .
p-value	The probability of obtaining a test statistic as extreme as observed, assuming H_0 is true.
Confidence interval	An interval that contains the parameter with a stated confidence level, $(1-\alpha)100\%$.
Correlation coefficient (r)	A measure of the strength and direction of the linear relationship between two variables, $r \in [-1, 1]$.
R^2	The proportion of variance in Y explained by X; $R^2 = r^2$.

Formulas (verbatim) — full sweep

Formula	Statement
Sturges	$k = 1 + 3.322 \log_{10}(n)$
Variance (pop)	$\sigma^2 = \Sigma(x-\mu)^2 / N$
Variance (sample)	$s^2 = \Sigma(x-\bar{x})^2 / (n-1)$
Bayes	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$
Var(X)	$E[X^2] - (E[X])^2$
Binomial	$P(X=x) = C(n,x) p^x (1-p)^{n-x}$
Poisson	$P(X=x) = e^{-1} \lambda^x / x!$
Standard error	$SE = \sigma / \sqrt{n}$
z-test (σ known)	$z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$
t-test (σ unknown)	$t = (\bar{x} - \mu) / (s/\sqrt{n})$
Chi-square	$\chi^2 = \Sigma (O - E)^2 / E$
CI for μ (σ known)	$\bar{x} \pm z_{\alpha/2} \times \sigma/\sqrt{n}$
CI for μ (σ unknown)	$\bar{x} \pm t_{\alpha/2} \times s/\sqrt{n}$
CI for proportion	$\hat{p} \pm z_{\alpha/2} \times \sqrt{(\hat{p}(1-\hat{p})/n)}$
Sample Size	$n = (z_{\alpha/2} \times \sigma / E)^2$
Regression slope	$b = \Sigma(x-\bar{x})(y-\bar{y}) / \Sigma(x-\bar{x})^2$
Regression intercept	$a = \bar{y} - b \bar{x}$
Correlation	$r = \Sigma(x-\bar{x})(y-\bar{y}) / \sqrt{(\Sigma(x-\bar{x})^2 \Sigma(y-\bar{y})^2)}$

Spearman	$\rho = 1 - 6\sum d^2 / (n(n^2-1))$
Probable Error	$P.E = 0.6745 (1 - r^2) / \sqrt{n}$
Markov steady state	$\pi P = \pi, \sum \pi_i = 1$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #1–#10 full sweep (canonical)
Numericals	Numerical-Book.md #46–#70 full sweep (descriptive → regression)

Books · Chapters · Media

Resource	Where
Weeks 1–13 packs	timed past papers
Devore	Ch 10 (supplementary) + online notes for Markov Chains
Module 4 supplement	03-Chapter-Breakdowns/04-... (Markov/random processes, ~5%)

Fear-Killer Pack Week 14 (verbatim)

Resources: all Weeks 1-8 packs • timed past papers

1. Timed full-length past paper (3 hrs, closed book) — START with the past paper, NOT with review.
2. Hypothesis testing: 4 complete problems (z , t , χ^2) — all from blank page.
3. Confidence intervals: 5 problems — retrieve formulas first.
4. Bayes: 3 problems — retrieve theorem from memory.
5. Regression: 2 complete problems — derive a , b from blank page.
6. Descriptive stats: compute mean, variance, skewness from raw data — closed book.
7. Markov Chain basics (30 min): state transition diagram, transition matrix, steady-state $\pi P = \pi$. Do NOT over-invest — ~5% weight.
8. Random Processes basics (20 min): stationary vs WSS, autocorrelation $R(t_1, t_2)$, ergodicity. Module-4 theory component.

Narrative — Week 14 (verbatim)

Topics: Cumulative review of Weeks 1-13; Past paper practice; Random Processes & Markov Chain basics (stationarity, autocorrelation of wide-sense stationary processes, ergodicity; Markov Chain & Queuing Theory) — ~5% weight, typically one short question

Resources: Devore Ch.10 (supplementary); online notes for Markov Chains

Retrieval protocol: No passive reading at all this week. Every minute is active retrieval. If you cannot retrieve a formula, that tells you exactly what to drill.

Sleep banking: Continue 9 hrs sleep (bedtime 22:00). Do NOT trade sleep for last-minute studying — it degrades recall by 15-20% per lost hour.

Practice: - Timed full-length past paper (3 hrs, closed book) — START with the past paper, NOT with review - Hypothesis testing: 4 complete problems (z , t , χ^2) — all from blank page - Confidence intervals: 5 problems — retrieve formulas first - Bayes: 3 problems — retrieve theorem from memory - Regression: 2 complete problems — derive a , b from blank page - Descriptive stats: compute mean, variance, skewness from raw data — closed book - **Markov Chain basics (30 min):** State transition diagram, transition matrix, steady-state probabilities $\pi P = \pi$. Do 2-3 problems to cover the expected 5% weight. Do NOT over-invest — this is a supplementary topic, not a core pillar. - **Random Processes basics (20 min):** Stationary vs wide-sense stationary, autocorrelation $R(t_1, t_2)$ of a WSS process, ergodicity/time averaging. 2-3 short problems — module-4 theory component. - **Spaced repetition:** Redrill any formula that took >30s to recall. Repeat until reflex-level.

Red flag: If hypothesis testing takes >20 minutes per problem, drill the 4-step procedure until it is automatic. This is your highest-weight topic.

Deliverable: Solved past paper + final formula sheet (1 page, double-sided)

Time budget: 8 hrs

- **P0 floor:** fear-killer pack pass (Q1 past paper + Q2 hypothesis testing) · def/formula skim (full W1–W13 sweep) · same-problem drill #59/#60/#64/#67
- **Same-problem drill target:** hypothesis testing < 20 min/problem; any formula recalled < 30 s; Markov steady-state $\pi P = \pi$ covered but not over-invested
- **Deliverable:** solved past paper + final formula sheet (1 page, double-sided)
- **Trap:** no passive reading — every minute active retrieval; START with the past paper, NOT with review; Markov/random processes ~5% — do NOT over-invest

W15 — FINAL EXAM WEEK · 23–29 Nov · Exam

Banner: Rotation PS (Mon+Tue) · Tier FINAL · No new deep study, no floor accrual. Ledger frozen during W15.

Sources

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 15	lines 297–315
Pack	Fear-Killer-Packs.md → Week 15	no pack
Week manifest	(W15 note at <code>weeks/PS-W14.md</code> L54)	reference

Fear-Killer Pack — Week 15 (verbatim)

Week 15: FINAL EXAM — No new pack. Execution only: numericals (hypothesis testing, CI, Bayes, regression, probability) → theory. Write the 4 steps explicitly for every hypothesis test.

Narrative — Week 15 (verbatim)

Focus: Execution. Do not learn anything new.

- Review 1-page formula sheet (final version prepared Week 14)
- Z-table: confirm you know how to read it (left tail, right tail, between values)
- Answer order: numericals (hypothesis testing, confidence intervals, Bayes, regression, probability) → theory (definitions, interpretations, comparisons)
- For hypothesis testing: write the 4 steps explicitly. Even if your calculation is wrong, correct structure earns 60% of marks

Final formula sheet checklist (must have): - Bayes' theorem - Z-test and t-test formulas - CI formulas (mean, proportion) - χ^2 formula - Regression formulas (a, b, r, R^2) - Binomial, Poisson, Normal PDF/PMF formulas - Markov Chain: $\pi P = \pi$, steady-state equation - Z-table and t-table critical values (common ones)

Exam-day stack (final)

- [] Numerical speed drills first — hypothesis testing, CI, Bayes, regression formulas from memory
- [] Blank-page retrieval — write the 4-step procedure from memory
- [] Theory last; flag stuck problems and move on

Notes

- Review the 1-page formula sheet (prepared Wk14). Sleep 8 h each night.
- Exam order: numericals → theory. Write the 4 hypothesis-testing steps explicitly.
- **Ledger frozen during Wk15** — exam window 30 Nov – 18 Dec is a separate phase (per `weeks/PS-W14.md` L54).